



AI Business Assistant for Spaza Shops & Township Entrepreneurs

Final Project — Build Plan (Day 1 to Finished Agent)

Bridging AI Solution Development Programme • Sabio Group

Team size	2 – 3 members
Duration	4 weeks (20 working days)
Built with	Python, LLMs, AI Agents, LangChain, RAG, Vector DB, Streamlit/Gradio
Goal	Help township small businesses manage stock, pricing, marketing & advice

The Vision

KasiBiz is an AI business assistant that helps spaza shop and township business owners manage stock, price their products, market their business, get simple financial advice, and understand how to register their business — all in plain language and local South African languages.

■ Team Roles (for 2–3 people)

Everyone codes, but each person owns one area so nothing falls through the cracks.

Role	Owner	Responsible for
AI / Backend Lead	Person 1	Agents, LLM logic, LangChain, prompt design
RAG / Data Lead	Person 2	Vector DB, documents, stock database, retrieval
Frontend / Integration Lead	Person 3*	Streamlit/Gradio UI, connecting everything, testing

**If you are only 2, combine Frontend with whoever has the lighter load that week.*

■ Tech Stack (everything you learnt this year)

Python	Core language for the whole project.
OpenAI API / LLM	The “brain” that understands and generates responses.
LangChain	Connects the LLM, tools and agents together.
AI Agents (multi-agent)	Split the work: a Stock Agent, a Marketing Agent, a Business-Advice Agent.
RAG + Vector DB (Chroma)	Answer questions from real documents (e.g. CIPC registration guides).
SQLite	A simple database to store stock/inventory.
Streamlit / Gradio	The user interface (you already have Streamlit experience).
GitHub	Version control and teamwork.

■ The Step-by-Step Plan (4 Weeks / 20 Days)

Week 1 — Foundation & Planning

Goal: Everyone agrees on what we are building and the base is set up.

Day 1 — Discovery & Scope

- As a team, agree on the core features (see MVP list).
- Write down what KasiBiz will and won't do (avoid scope creep).
- Decide who owns which role.

Day 2 — Design the User Journey

- Sketch how a shop owner talks to KasiBiz (“add stock”, “write me a WhatsApp advert”).
- Draw a simple flow diagram of the agents and features.

Day 3 — Environment & GitHub Setup

- Create the GitHub repo and add all team members.
- Set up the Python virtual environment and install libraries (LangChain, OpenAI, Chroma, Streamlit).
- Create a .env file for API keys and a clean folder structure.

Day 4 — “Hello World” LLM Connection

- Get a basic script talking to the OpenAI API.
- Test a simple prompt and response so everyone's setup works.

Day 5 — Gather Data & Documents

- Collect RAG documents (CIPC registration steps, budgeting tips, SARS basics).
- Prepare a sample stock list (product name, price, quantity).

Week 2 — Core Features (The Engine)

Goal: Build the main brains of KasiBiz, feature by feature.

Day 6 — Stock Database

- Build the SQLite database for stock (add, view, update, delete products).
- Test adding and reading products with Python.

Day 7 — Stock Agent

- Create the Stock Agent that answers “What's low in stock?” and “What should I reorder?”.
- Add logic: if quantity is below a set level, suggest a reorder.

Day 8 — Pricing Helper

- Suggest selling prices (cost price + markup) and calculate profit.
- Let the LLM explain the pricing in simple terms.

Day 9 — Marketing Agent

- Create the Marketing Agent that generates WhatsApp adverts, social captions and poster text.

Day 10 — Business Advice Agent + RAG (Part 1)

- Load the documents into the Chroma Vector Database (embeddings).

- Build the retriever so the system can search the documents.

Week 3 — Bring It Together (Multi-Agent + RAG)

Goal: Connect all the pieces into one smart assistant.

Day 11 — Business Advice Agent + RAG (Part 2)

- Connect the RAG retriever to an agent so it answers “How do I register with CIPC?” using real documents and cites the source.

Day 12 — The Router / Orchestrator

- Build the main agent that decides which sub-agent handles each request — your intent classification, just like at Sabio.

Day 13 — Multilingual Support

- Add replies in local languages (isiZulu, isiXhosa, Afrikaans, etc.).
- Test that answers stay accurate when translated.

Day 14 — Memory & Context

- Add conversation memory so KasiBiz remembers context within a chat (e.g. the product being discussed).

Day 15 — Integration Day

- Connect all agents and features through the router.
- Full internal test: can one chat move between stock, pricing, marketing and advice smoothly?

Week 4 — Interface, Testing & Demo

Goal: Make it usable, reliable and presentable.

Day 16 — Build the Interface

- Build the Streamlit/Gradio UI: a chat window, a stock view and clear buttons/menus.

Day 17 — Connect UI to Backend

- Wire the interface to all agents so everything works from the screen (no code needed to use it).

Day 18 — Testing & Red-Teaming

- Test like real shop owners would — weird inputs, wrong spellings, tricky questions (use the Sabio red-team mindset).
- Log bugs and fix them.

Day 19 — Polish & Documentation

- Improve design, error messages and responses.
- Write a short README and a simple user guide.

Day 20 — Final Demo Prep

- Prepare a demo script and presentation slides.
- Do a full practice run-through as a team.
- ■ KasiBiz is finished and ready to present!

■ MVP (Must-Have) vs. Stretch Goals

Finish the MVP first, then add extras if there is time.

MVP — build these first	Stretch goals — if time allows
1. Stock tracking + reorder suggestions	• Multilingual replies
2. Pricing & profit helper	• Voice input / output
3. Marketing message generator	• Sales tracking & simple reports
4. Business advice via RAG (CIPC/budgeting)	• WhatsApp integration
5. Simple chat interface	

■ Tips to Keep the Team On Track

- **Daily 10-min stand-up:** each person says what they did, what's next, and any blockers.
- **Commit to GitHub often** so no work is lost and everyone stays in sync.
- **Test as you build** — don't leave all testing for the end.
- **Keep it simple first** — a working small version beats an unfinished big one.

KasiBiz — empowering local business, one conversation at a time. ■■